

STROBE Statement Checklist

Study: Early physiological phenotypes and outcomes in critically ill adults with non-traumatic subarachnoid hemorrhage

Design: Retrospective cohort study using MIMIC-IV 3.1 with external validation in eICU-CRD.

This checklist is prepared for journal submission and should be reviewed by all authors before upload.

Item	STROBE recommendation	Manuscript location	Status
1a	Indicate the study design in the title or abstract	Title; Structured Abstract, Methods	Addressed
1b	Provide an informative and balanced summary	Structured Abstract	Addressed
2	Explain the scientific background and rationale	Introduction	Addressed
3	State specific objectives and hypotheses	Introduction, final paragraph	Addressed
4	Present key elements of study design early	Methods, Study design and data sources	Addressed
5	Describe setting, locations and relevant dates	Methods, Study design and data sources; ESM	Addressed
6a	Give eligibility criteria and methods of participant selection	Methods, Cohort; ESM Tables 1–3	Addressed
6b	For matched studies, give matching criteria	Not applicable	Not applicable
7	Define outcomes, exposures, predictors, confounders and effect modifiers	Methods; ESM Tables 4–6	Addressed
8	For each variable, give sources of data and details of assessment	Methods; ESM variable mapping	Addressed
9	Describe efforts to address sources of bias	Methods; Discussion limitations; sensitivity analyses	Addressed
10	Explain study size	Cohort flow; ESM Tables 1–2	Addressed
11	Explain quantitative variables and grouping	Methods, Variables and preprocessing; ESM Tables 4–6	Addressed

Item	STROBE recommendation	Manuscript location	Status
12a	Describe all statistical methods, including confounding control	Methods, Outcomes and statistical analysis	Addressed
12b	Describe methods used to examine subgroups/interactions	Methods, sensitivity analyses; ESM Table 12	Addressed
12c	Explain how missing data were addressed	Methods; ESM Table 5	Addressed
12d	Explain cohort follow-up or matching methods	Methods; ESM cohort-flow tables	Addressed
12e	Describe sensitivity analyses	Methods; ESM Table 12	Addressed
13a	Report numbers at each stage of study	Figure 1; ESM Tables 1–2	Addressed
13b	Give reasons for non-participation/exclusion	Figure 1; Methods; ESM Tables 1–2	Addressed
13c	Consider use of a flow diagram	Figure 1	Addressed
14a	Give characteristics of study participants	Results; ESM Table 7	Addressed
14b	Indicate number with missing data for each variable	ESM Table 5	Addressed
14c	Summarize follow-up time	ESM Tables 7 and 11 include length of stay; Cox model uses hospital stay	Partly addressed
15	Report outcome event numbers or summary measures	Results; ESM Tables 7, 11, 12	Addressed
16a	Give unadjusted and adjusted estimates with precision	Results; ESM Tables 9–10	Addressed
16b	Report category boundaries for categorized continuous variables	Methods; ESM variable definitions	Addressed
16c	Consider translating relative risk into absolute risk	Results include absolute mortality gradients	Addressed
17	Report other analyses, including sensitivity analyses	Results; ESM Table 12; ESM Figs. 1–7	Addressed
18	Summarize key results with reference to objectives	Discussion, first paragraph	Addressed
19	Discuss limitations and direction/magnitude of potential bias	Discussion, limitations	Addressed
20	Give cautious overall interpretation	Discussion and Conclusions	Addressed
21	Discuss generalisability	Discussion, limitations	Addressed
22		Declarations	Needs author completion

Item	STROBE recommendation	Manuscript location	Status
	Give source of funding and role of funders		

Author Completion Items

- Confirm local ethics/IRB wording for analysis of de-identified public databases.
- Complete funding statement.
- Complete conflicts of interest.
- Complete author contributions.
- Confirm final code and data availability statements.
- Export final references from a reference manager and verify all bibliographic details.